

IDSEA.IN VPS Installation Guide

Complete Step-by-Step Deployment on Ubuntu 24.04

With Animitra + VetMechPharma already running on the same VPS

Domain	idsea.in
Backend Port	8003
VPS OS	Ubuntu 24.04
Database	MongoDB 7.0 (already installed)
Backend	Python 3.11 + FastAPI
Frontend	React (static build served by Nginx)
Estimated Time	20-30 minutes

Step 0: SSH into Your VPS

Open terminal on your computer and connect to your VPS:

```
ssh root@YOUR_VPS_IP
```

Replace `YOUR_VPS_IP` with your actual VPS IP address. Example: `ssh root@103.45.67.89`

Step 1: Check Existing Ports (Avoid Conflicts)

Before anything, check which ports Animitra and VetMechPharma are using:

```
sudo ss -tlnp | grep -E '800[0-9]|300[0-9]|500[0-9]'
```

You'll see output showing which ports are in use. We will use **port 8003** for IDSEA.

```
sudo supervisorctl status
```

This shows all running projects. Note them down — they should keep running.

If port 8003 is already in use, use 8004 instead everywhere in this guide.

Step 2: Check Existing Nginx Sites

```
ls -la /etc/nginx/sites-enabled/
```

You should see config files for animitra and vetmechpharma. We will ADD a new file for idsea.in — we will NOT touch the others.

Step 3: Create IDSEA Directory

```
sudo mkdir -p /var/www/idsea sudo mkdir -p /var/log/idsea cd /var/www/idsea
```

Step 4: Clone Code from GitHub

```
cd /var/www/idsea sudo git clone YOUR_GITHUB_REPO .
```

Replace `YOUR_GITHUB_REPO` with your actual GitHub repository URL. The dot (.) at the end means clone INTO current directory.

If it asks for credentials, use a GitHub Personal Access Token (GitHub > Settings > Developer settings > Personal access tokens).

After cloning, verify:

```
ls -la # You should see: backend/ frontend/ deploy/ ...
```

Step 5: Setup Backend (Python)

```
cd /var/www/idsea/backend python3 -m venv venv source venv/bin/activate pip install --upgrade pip pip install -r requirements.txt deactivate
```

```
mkdir -p /var/www/idsea/backend/uploads sudo chown -R www-data:www-data /var/www/idsea/backend/uploads
```

Step 6: Create Backend .env File

First, generate a random JWT secret key:

```
python3 -c "import secrets; print(secrets.token_hex(32))"
```

COPY the output — you'll paste it in the .env file below.

Create the .env file:

```
sudo nano /var/www/idsea/backend/.env
```

Paste this content (edit the lines marked with arrows):

MONGO_URL	mongodb://localhost:27017
DB_NAME	idsea_db
CORS_ORIGINS	https://idsea.in,https://www.idsea.in
JWT_SECRET_KEY	PASTE_YOUR_GENERATED_SECRET <<<
SMTP_HOST	smtp.gmail.com
SMTP_PORT	587
SMTP_USER	your-email@gmail.com <<<
SMTP_PASS	your-gmail-app-password <<<
FROM_EMAIL	noreply@idsea.in
RAZORPAY_KEY_ID	(leave blank for now)
RAZORPAY_KEY_SECRET	(leave blank for now)

Save: Press Ctrl+O, then Enter, then Ctrl+X to exit nano.

Gmail SMTP: Go to <https://myaccount.google.com/apppasswords> to generate an App Password. Don't use your real Gmail password.

Razorpay: You can set keys later from Admin Panel. Leave blank for now.

Step 7: Create Frontend .env File

```
sudo nano /var/www/idsea/frontend/.env
```

Paste this content:

```
REACT_APP_BACKEND_URL=https://idsea.in REACT_APP_RAZORPAY_KEY_ID=
```

Save: Ctrl+O, Enter, Ctrl+X

Step 8: Build Frontend

```
cd /var/www/idsea/frontend yarn install yarn build
```

This takes 1-2 minutes. When done, verify:

```
ls -la build/ # Should see index.html and static/ folder (~6-7MB)
```

Step 9: Setup Supervisor (Backend Process)

This keeps the IDSEA backend running and auto-restarts if it crashes.

```
sudo cp /var/www/idsea/deploy/supervisor-idsea.conf /etc/supervisor/conf.d/idsea.conf
```

Or create manually:

```
sudo nano /etc/supervisor/conf.d/idsea.conf
```

Content of supervisor config:

```
[program:idsea-backend] command=/var/www/idsea/backend/venv/bin/uvicorn server:app
--host 0.0.0.0 --port 8003 --workers 2 --log-level info
directory=/var/www/idsea/backend user=www-data autostart=true autorestart=true
startretries=5 stderr_logfile=/var/log/idsea/backend.err.log
stdout_logfile=/var/log/idsea/backend.out.log stderr_logfile_maxbytes=10MB
stdout_logfile_maxbytes=10MB environment=PATH="/var/www/idsea/backend/venv/bin"
```

Load and start:

```
sudo supervisorctl reread sudo supervisorctl update sudo supervisorctl start
idsea-backend
```

Verify ALL projects are running:

```
sudo supervisorctl status
```

You should see idsea-backend RUNNING alongside your other projects.

```
sleep 3 curl http://localhost:8003/api/public/events
```

If you see JSON output, the backend works! If error, check: `sudo tail -30 /var/log/idsea/backend.err.log`

Step 10: Setup Nginx (Web Server)

This step adds a NEW config file. It does NOT touch Animitra or VetMechPharma configs.

```
sudo cp /var/www/idsea/deploy/nginx-idsea.conf /etc/nginx/sites-available/idsea.in
```

Or create manually:

```
sudo nano /etc/nginx/sites-available/idsea.in
```

Full Nginx config content (port 8003, domain idsea.in):

```
server { listen 80; server_name idsea.in www.idsea.in; client_max_body_size 25M; root
/var/www/idsea/frontend/build; index index.html; gzip on; gzip_vary on;
gzip_min_length 1024; gzip_types text/plain text/css application/json
application/javascript text/xml application/xml text/javascript image/svg+xml;
location /api/ { proxy_pass http://127.0.0.1:8003/api/; proxy_http_version 1.1;
proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr;
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for; proxy_set_header
X-Forwarded-Proto $scheme; proxy_read_timeout 300s; } location /uploads/ { alias
/var/www/idsea/backend/uploads/; expires 30d; add_header Cache-Control "public,
immutable"; } location ~* \.(js|css|png|jpg|jpeg|gif|ico|svg| woff|woff2|ttf|eot)$ {
expires 1y; add_header Cache-Control "public, immutable"; access_log off; } location
/ { try_files $uri $uri/ /index.html; } add_header X-Frame-Options "SAMEORIGIN"
always; add_header X-Content-Type-Options "nosniff" always; }
```

Enable the site and test:

```
sudo ln -s /etc/nginx/sites-available/idsea.in /etc/nginx/sites-enabled/idsea.in
```

```
sudo nginx -t
```

IMPORTANT: If nginx -t shows any error, DO NOT reload nginx! Fix the error first. If you reload with an error, it will break ALL your sites including Animitra and VetMechPharma.

If test passes (shows 'syntax is ok'), reload:

```
sudo systemctl reload nginx
```

Step 11: Point Domain DNS to VPS

Go to your domain registrar (GoDaddy, Namecheap, etc.) and set these DNS records:

Type	Host	Value	TTL
A	@	YOUR_VPS_IP	600
A	www	YOUR_VPS_IP	600

Replace *YOUR_VPS_IP* with your actual VPS IP. DNS takes 5-30 minutes to propagate.

Check if DNS is pointing correctly:

```
dig idsea.in +short # Should show your VPS IP
```

Step 12: Install SSL Certificate (HTTPS)

Only do this AFTER DNS is pointing to your VPS (Step 11).

```
sudo certbot --nginx -d idsea.in -d www.idsea.in
```

Certbot will ask:

- Enter email: **your-email@gmail.com**
- Agree to terms: **Y**
- Share email: **N**
- Redirect HTTP to HTTPS: **2 (Yes)**

Verify HTTPS works:

```
curl -I https://idsea.in # Should return HTTP/2 200
```

Step 13: Verify Everything Works

```
echo '=== All Services ===' sudo supervisorctl status sudo systemctl status nginx |  
grep Active sudo systemctl status mongod | grep Active curl -s  
https://idsea.in/api/public/events | head -c 100
```

Open in your browser:

Website	https://idsea.in
Admin Panel	https://idsea.in/admin
Login Email	admin@idsea.org
Login Password	Admin@123

IMPORTANT: Change the admin password after first login!

Step 14: Setup Daily Backup (Recommended)

```
sudo mkdir -p /var/backups/idsea  
  
(sudo crontab -l 2>/dev/null; echo "0 2 * * * mongodump --db idsea_db --out  
/var/backups/idsea/\$(date +%Y%m%d) --quiet") | sudo crontab -
```

This backs up the MongoDB database daily at 2 AM.

Useful Commands (Bookmark This Page)

What	Command
Check IDSEA status	<code>sudo supervisorctl status idsea-backend</code>
Restart IDSEA backend	<code>sudo supervisorctl restart idsea-backend</code>
View error logs	<code>sudo tail -50 /var/log/idsea/backend.err.log</code>
View output logs	<code>sudo tail -50 /var/log/idsea/backend.out.log</code>
Reload Nginx	<code>sudo systemctl reload nginx</code>
Check ALL projects	<code>sudo supervisorctl status</code>
MongoDB shell	<code>mongosh idsea_db</code>
Renew SSL	<code>sudo certbot renew</code>

How to Update IDSEA (After Code Changes)

```
cd /var/www/idsea sudo git pull origin main # If backend changed: cd backend source
venv/bin/activate pip install -r requirements.txt deactivate sudo supervisorctl
restart idsea-backend # If frontend changed: cd /var/www/idsea/frontend yarn install
yarn build sudo systemctl reload nginx
```

Troubleshooting

Problem	Solution
502 Bad Gateway	Backend not running. Run: <code>sudo supervisorctl restart idsea-backend</code> <code>sudo tail -30 /var/log/idsea/backend.err.log</code>
Blank white page	Frontend build missing. Run: <code>cd /var/www/idsea/frontend && yarn build</code>
SSL certificate error	Re-run: <code>sudo certbot --nginx -d idsea.in -d www.idsea.in</code>
Upload fails (413)	Check nginx has: <code>client_max_body_size 25M;</code>
Other projects broke	Check: <code>sudo nginx -t</code> <code>ls /etc/nginx/sites-enabled/</code> Ensure their config files still exist